Andre (Jianyou) Wang

jiw101@ucsd.edu, Google Scholar

Education

Ph.D., Computer Science and Engineering, University of California, San Diego **2021–2026**

- Advisors: Leon Bergen & Ramamohan Paturi. Expected graduation Oct 2026.
- Specialty: LLM Reasoning & Post-Training; Reinforcement Fine-Tuning; Real-World Forecasting; Long-Context Modeling; Counterfactual Reasoning; AI4Science & DeepResearch Agents; Retrieval Augmented Generation; Instruction-Following Information Retrieval; Synthetic Data; Trustworthy Large-Scale Evaluation and Benchmarking.

B.S., Mathematics, Computer Science, Statistics, *with Highest Distinction*, Duke University **2017–2021**

- Advisors: Cynthia Rudin & Vahid Tarokh. Julia Dale Prize (top three first-year mathematics students).

Selected Research Projects

Real-World Forecasting, Counterfactual Post-Training & Benchmarks — *DeepImagine, CT Open*

- **DeepImagine:** Reframed clinical-trial outcome forecasting as stepwise counterfactual imagination. Modeled an intractable joint prediction as a Markov chain of single-factor perturbations. Trained specialized Gemma-3-12B operators on natural and synthetic counterfactuals via SFT, RFT and GRPO. Under a calibration assumption, proved multi-anchor aggregation converges to the true conditional. Empirically observed variance-reduction signature. Our model matched GPT-5 and outperformed RAG and traditional baselines.
- **CT Open:** Built the first live, contamination-resistant benchmark for prospective clinical-trial forecasting: an agentic pipeline (search, scraping, multi-round verification) with time-stamped cutoffs, validated by external-database checks and physician-scientist audit. Hosting the prediction challenge at Website.

Reinforcement Learning & Self-Improving Curricula — *Question-begets-Question*

- Built a self-evolving RL curriculum that teaches a model a skill it hasn't mastered with *no* ground-truth reasoning traces: each round self-evaluates, synthesize new problems, and fine-tunes with GRPO.
- Overcame an apparent RL plateau under fixed data/compute (5.6% $\rightarrow$ 16.5% pass@1, no saturation after 20 rounds). Analyzed the mechanism: seeding only from "mostly-solved" problems succeeds while harder or mixed seeds stay at baseline, because boundary-level seeds keep GRPO groups in a high-variance regime whereas too-hard seeds zero out the group-normalized advantage. Gains transfer to unseen hard problems.

Retrieval Augmented Generation & Long-Context Modeling — *Single-Pass Document Scanning*

- Framed long-document Q&A as RAG: a cheap scanner retrieves key sentences to feed the LLM generator.
- Turned a pretrained Mamba-2 into a single-pass document scanner: replaced the LM head with a binary classification head scoring each sentence in one linear-time pass; the discriminative design beat generative fine-tuning of the same backbone by $+26.5$ points $(33.5 \rightarrow 60.0)$.
- A 130M scanner beat 7B embedding models (NV-Embed-v2, GritLM) at $\sim$70$\times$ fewer TFLOPs; using 1M carefully designed synthetic training examples. Observed length generalization exceeding 256,000 tokens.

Information Retrieval & Ranking — *IR2, BIRCO*

- **IR2**: Designed a synthetic query-generation method with controllable complexity that beats Promptagator across four architectures, E5-v2, SimCSE, RoBERTa, SPECTER-v2, trained via contrastive fine-tuning.
- **BIRCO:** Introduced the first instruction-following information retrieval benchmark for complex user query.

Trustworthy Evaluation at Scale — *DORIS-MAE*

- **DORIS-MAE:** Built statistically validated LLM-as-expert-annotator pipelines (prespecified hypothesis tests, hierarchical bootstrap) matching expert panels of PhDs and MDs; annotated 165,000+ document–aspect pairs in $\sim$24 hours for under \$150, which is a 500$\times$ time and 1,000$\times$ cost reduction at no loss in quality.
- **RoBBR:** Decomposed risk-of-bias assessment into separate retrieval and reasoning subtasks, then ran a two-stage ablation swapping each component independently, isolating retrieval as the bottleneck and showing GPT-4o and Sonnet-3.5 have distinct strengths (retrieval vs. reasoning).

Internship & Industry Collaboration

Elsevier — ScienceDirect & Scopus AI teams, External Research Collaborator **2024–2026**

- Co-develop the Research Map knowledge graph, EvidenceBench and CT Open evaluation platforms.
- **Research Map:** Constructing a temporal knowledge graph of 40M biomedical papers; identifying subgraphs tracing how experimental results, relevant ideas, and technological advances drive hypotheses into clinical trials; training a DeepResearch agent to learn these evolutionary dynamics and predict future discoveries.

First-Author Publications & Preprints

1. Single-Pass Document Scanning for Question Answering[†]. *COLM 2025*, **Oral Spotlight (top 2%)**. Link.
2. Scientific Document Retrieval using Multi-Level Aspect-Based Queries (DORIS-MAE). *NeurIPS 2023* (Datasets & Benchmarks). Link.
3. DeepImagine: Clinical Trial Outcome Prediction via Stepwise Local Counterfactual Imaginations. *Preprint*. Link
4. Question Begets Question: Self-Evolving Curriculum for Reinforcement Fine-Tuning on Competition Mathematics. *Preprint*. Link
5. CT Open: An Open-Access, Uncontaminated, Live Platform for Clinical Trial Outcome Prediction. *COLM 2026*. Link. Website.
6. EvidenceBench: A Benchmark for Extracting Evidence from Biomedical Papers. *COLM 2025*. Link.
7. RoBBR: Measuring Risk of Bias in Biomedical Reports. *EMNLP 2025*. Link.
8. BIRCO: A Benchmark of Information Retrieval Tasks with Complex Objectives[†]. *Preprint*. Link.
9. IR2: Information Regularization for Information Retrieval. *LREC-COLING 2024* (Oral). Link.
10. Automated Limerick Poetry Generation with GPT-2. *Transactions of the ACL (TACL) 2021*. Link.
11. Speech Emotion Recognition with Dual-Sequence LSTM Architecture. *ICASSP 2020*. Link.

[†] Indicates equal contribution (co-first author).

Skills

- **ML/Infra:** LLM & state-space (Mamba) training/fine-tuning, LoRA/PEFT, PyTorch DDP, multi-GPU H100 clusters, synthetic-data pipelines, contrastive learning, RAG, long-context modeling, reinforcement learning.
- **Programming:** Python, PyTorch, Unix, SQL, Bash. **Languages:** English (native), Chinese (native).